ANANYA SHARMA

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Research interests: biomicrofluidics — microscale transport, cell mechanics and label-free cell separation for point-of-care diagnostics

EDUCATION

MIT Art, Design and Technology University (MIT School of Bioengineering Sciences & Research) Pune, India
B.Tech. Bioengineering • CGPA 9.80/10 • Rank 2 in cohort Aug. 2023 – Jun. 2027 (expected)
Relevant coursework: Applied Mathematics, Basic Mechanical Engineering, Biophysics, Microfluidics, Sensor Technology, Nanotechnology, Biomaterials

RESEARCH

Independent Research: Microscale Transport Modelling (sole author) Pune, India
MIT-ADT University 2026 – Present

Project Title: Mass Transport and Capture in Microchannel Surface Sensors

- Formulated and solved a convection–diffusion–reaction model of how much of a finite sample a microchannel surface sensor can capture.
- Showed capture peaks at flow rate Q* = V/T and identified a dimensionless kinetic capture ceiling set by binding rate, receptor density and sensor area.
- Derived a universal capture map and closed-form design rules accurate to within −2.2 to +1.4% of full numerical simulations (under review, Microfluidics and Nanofluidics; Research Square preprint).

Undergraduate Researcher, Microfluidic Devices | Advisor: Dr. Sulaxana Pandey Pune, India
MIT School of Bioengineering Sciences & Research 2024 – Present

- **Density-based blood-cell screening** (2025–26): an integrated magnetic-levitation chip failed on the bench, so built two separate prototypes, a ₹300 thread-driven centrifuge and a ₹2,000 tabletop centrifuge with an NdFeB magnetically levitated rotor, both electricity-free, separating blood in heparinized capillaries held in a custom PDMS chip; a smartphone with a microfluidic lens images the sample and an AI app returns blood counts. *Both tested and demonstrated*; SIH 2025 Institute Round winner; Top 10, Health Hack 2026.
- **Lab-on-chip microheater** (2025; co-advised by Dr. Navya Sethu): Arduino PID-controlled nichrome heater with thermocouple feedback, holding PDMS chips within ±0.5 °C; thermal characterisation, calibration and insulation testing. *Complete; manuscript in preparation.*
- **Rabies diagnostic chip** (2026–): finger-prick plasma separation, anti-nucleoprotein capture and gold-nanoparticle colorimetric readout scored by a smartphone CNN. *In development.*
- **Quantum-dot contact lens** (2025–): Sappan-wood quantum dots in PDMS for sustained, diffusive ocular drug release. *Lenses fabricated; drug loading in progress*; 1st place poster, ICRTB 2026.

Research Intern | Advisor: Prof. Sai Siva Gorthi Bangalore, India
Indian Institute of Science (IISc), Optics & Microfluidics Instrumentation Lab Jun. 2026 – Aug. 2026

Project Title: Plasma Biomarker Immunoassays for Alzheimer's Disease (with Centre for Brain Research & IBRO)

- Developed sandwich ELISAs for p-tau217 and NfL at pg/mL levels in human plasma, the ground-truth reference for an affordable point-of-care Alzheimer's biosensor.
- Designed a glutaraldehyde-free surface immobilisation route (plasma activation → EDC/sulfo-NHS coupling → ethanolamine quench) on polystyrene; diagnosed and reduced non-specific binding in plasma through blocking and controls.
- Built a prototype p-tau217 assay cartridge and performed optical-microfluidic imaging and image analysis toward a glass-cartridge readout.

Independent Computational Research: Validation of Biomedical Models Pune, India
MIT-ADT University 2026 – Present

- **Wearable blood lactate** (sole author): in 35 participants, an exercise-stage-only baseline gave MAE 0.966 mmol/L and AUC 0.945 vs 1.366 and 0.769 for heart rate alone, so heart-rate baselines overstate sensing gains (under review, Physiological Measurement).
- **EEG workload** (first author, equal contribution): subject-level conformal calibration restored coverage for unseen users (0.787 → 0.908), replicated on an external dataset (under review).
- **Reproducibility** (sole author): a single random split let a worse model beat a better one in 56% of 500 splits (Research Square preprint); built OrganoidAI Map, a database of 245 ML studies on organoids and organs-on-chips (Zenodo).

PAPERS AND PRESENTATIONS

Journal manuscripts under review

Sharma, A. *How much of a finite sample can a microchannel surface sensor capture? Optimal flow rate, universal capture map and closed-form design rules.* Microfluidics and Nanofluidics. Under review, submitted Oct. 2026. Preprint: Research Square, doi:10.21203/rs.3.rs-11390083/v1

Sharma, A. *Exercise stage explains most of the accuracy of blood-lactate estimation in graded exercise tests.* Physiological Measurement. Under review, submitted Sep. 2026. Code: github.com/ananya484/workloadcal

Sharma, A.*, Kale, A.*, & Pattanashetti, A.* (*equal contribution). *Subject-level conformal calibration for uncertainty-aware mental-workload detection on consumer-grade EEG: leave-one-subject-out evaluation with external validation.* Biomedical Signal Processing and Control. Under review, submitted Oct. 2026. Code: github.com/ananya484/eeg-workload-gradient-boosting

Preprints and datasets

Sharma, A. *The declared winner of a clinical prediction model comparison is often decided by the random split: a simulation study.* Research Square. Preprint, 2026. doi:10.21203/rs.3.rs-10794678/v1

Sharma, A. *OrganoidAI Map: a coded database of machine-learning studies on organoids and organs-on-chips (245 studies).* Zenodo. Database, v1.2.0, 2026. doi:10.5281/zenodo.23835346

Conference presentations

Sharma, A., Tawade, A., Sankpal, A., Sethu, N., & Pandey, S. *Development of a low-cost temperature-controlled microheater for lab-on-chip platforms.* 9th International Conference on Recent Trends in Bioengineering (ICRTB 2026), Pune, India. Poster PP-30 (abstract p. 72): January 2026.

Sharma, A., Tawade, A., & Pandey, S. *Smart contact lens technologies for ocular diagnostics.* ICRTB 2026, Pune, India. Poster PP-29 (abstract p. 71): January 2026.

In preparation

Sharma, A., et al. *Low-cost temperature-controlled microheater for PDMS microfluidic lab-on-chip systems.*

Sharma, A., et al. *PDMS–quantum-dot contact lens for sustained ocular drug delivery.*

SKILLS

Transport & Modelling	Microscale fluid mechanics and mass transport (convection–diffusion–reaction, dimensionless scaling, closed-form approximations), finite-difference and FEM (COMSOL Multiphysics) simulation, Monte Carlo studies
Microfabrication	Soft lithography, photolithography, PDMS device fabrication, microfluidic chip design
Hardware & Design	PID control, thermocouple feedback, Arduino, ESP, IoT; SolidWorks, AutoCAD
Programming	Python (NumPy, SciPy, scikit-learn), C++, Java, SQL; CNNs, gradient boosting, image processing
Lab & Characterisation	Sandwich ELISA, EDC/sulfo-NHS surface coupling, 4PL standard curves, microscopy, DLS, zeta potential, FTIR, contact angle

AWARDS & HONORS

Academic Merit Medals, MIT-ADT University (Silver Yr 1, Gold Yr 2, Silver Yr 3) 2024–2026
SURE Fellowship, IIT Hyderabad (funded Summer Undergraduate Research Exchange; selected from 50,000+ applicants) 2026
1st Place, ICRTB 2026 Poster Presentation (Biomedical Theme) 2026
Top 10, Health Hack 2026 (in association with Johns Hopkins University, USA) 2026
Smart India Hackathon, Institute Round: Winner (Hardware, 500+ teams, 2025); 2nd (Smart Automation, 1,500+ teams, team lead, 2026); 4th (Software, 2024) 2024–2026

ADDITIONAL EXPERIENCE & LEADERSHIP

Research Intern, synthetic microbial consortia for drought resilience (growth and co-culture assays), MIT SBSR Pune May 2026 – Present
MEDHA Medical Device Hackathon (with IIT Bombay): smart neonatal incubator with jaundice detection Nov. 2025
Community Service Director, Rotaract Club of Pune Camp Pioneers Jul. 2025 – Jun. 2026
Volunteer, Robin Hood Army, Pune Jun. 2024 – Jul. 2025

OTHER

Languages	English (C2), Hindi (native), Marathi, French (A2), German (basic)
Interests	Bharatanatyam (10 years, diploma; pursuing Bachelor's in the form), badminton, handball, painting